JINGJING LU

Plasma Control · Data-Driven Modeling · AI for Science

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RESEARCH PROFILE

Doctoral researcher at the University of Science and Technology of China (USTC), conducting research at the Institute of Plasma Physics, Chinese Academy of Sciences (ASIPP). Research focuses on machine learning for EAST plasma equilibrium reconstruction, feedforward coil-current trajectory design, diagnostic-data reconstruction, and uncertainty-aware prediction. First-author research published in *AIP Advances* and presented at international conferences.

EDUCATION

University of Science and Technology of China / ASIPP 2023 – present
Doctoral studies in Energy and Power Engineering **Expected graduation: June 2027**

University of Science and Technology of China / ASIPP 2020 – 2023
Master's degree, Energy and Power Engineering

Fuyang Normal University 2015 – 2019
Bachelor's degree, Physics

RESEARCH EXPERIENCE

Fast Plasma Equilibrium Reconstruction on EAST

- Developed a deep-learning approach mapping magnetic measurements to two-dimensional poloidal magnetic flux, supervised by offline EFIT reconstructions.
- Used automated hyperparameter optimization and evaluated predicted flux surfaces, the last closed flux surface, and normalized internal inductance against EFIT. Published first-author results in *AIP Advances* (2023).

Physics-Informed Imitation Learning for Feedforward Coil-Current Design

- Investigate physics-informed imitation learning and Seq2Seq Transformer methods for EAST feedforward coil-current trajectory design, focusing on temporal modeling and physical constraints.
- First-author manuscript submitted to *Communications Physics*; invited presentation at AAPPS-DPP 2026 and oral presentation at an IAEA technical meeting.

Diagnostic Reconstruction and Uncertainty-Aware Prediction — Collaborative Research

- Co-authored research on reconstructing missing electron temperature using time series extrinsic regression (*Nuclear Fusion*, 2025; third author).
- Co-authored research on Bayesian neural networks for energy confinement time prediction and uncertainty quantification (*Nuclear Fusion*, 2025; fifth author).

TECHNICAL SKILLS

Programming and ML: Python, PyTorch; deep neural networks, time-series modeling, Seq2Seq Transformer.

Domain tools: EFIT; EAST magnetic-equilibrium and diagnostic-data analysis; feedforward coil-current design.

Languages: Chinese; English (CET-6).

SELECTED PUBLICATIONS

First-author papers (including submitted work)

[1] **J. Lu**, Y. Hu, N. Xiang, and Y. Sun. Fast equilibrium reconstruction by deep learning on EAST tokamak. *AIP Advances* **13**, 075007 (2023). DOI: 10.1063/5.0152318.

[2] **J. Lu** et al. Physics-informed imitation learning for feedforward coil-current trajectory design on EAST. Submitted to *Communications Physics*.

Co-authored papers

[3] M. Wang, C. Wan, **J. Lu** et al. Time series extrinsic regression for reconstructing missing electron temperature in tokamak. *Nuclear Fusion* **65**, 076008 (2025). DOI: 10.1088/1741-4326/addb5f.

[4] E. Gao, C. Wan, Y. Hu, M. Wang, **J. Lu** et al. Bayesian neural networks for predicting tokamak energy confinement time with uncertainty quantification. *Nuclear Fusion* **65**, 084001 (2025). DOI: 10.1088/1741-4326/ade8fd.

[5] Y.-Y. Liu, D.-K. Yang, K. Zhu, **J.-J. Lu** et al. T-DDFM: an advanced denoising diffusion fusion model for high-precision neutron imaging. *Nuclear Science and Techniques* **37**, 231 (2026). DOI: 10.1007/s41365-026-02054-6.

PATENT

Youwen Sun, **Jingjing Lu**, et al. (second inventor). Method and system for feedforward current design of tokamak poloidal-field coils. Granted Chinese invention patent ZL202611178059.9; publication No. CN122735600B; granted 30 October 2026. (Title translated from Chinese.)

SELECTED PRESENTATIONS

Invited talk 11 October 2026
Feedforward poloidal field coil current design on EAST via behavioral cloning.
10th Asia-Pacific Conference on Plasma Physics (AAPPS-DPP 2026), Busan, Republic of Korea

Oral presentation 9 September 2025
Data processing for imitation learning in EAST feedforward current design.
6th IAEA Technical Meeting on Fusion Data Processing, Validation and Analysis, Shanghai, China

Oral presentation 27 October 2023
Fast equilibrium reconstruction on the EAST tokamak using deep learning.
11th Conference on Magnetic Confinement Fusion Theory and Simulation, Chengdu, China. (Title translated from Chinese.)

HONORS

USTC academic scholarships: first class (twice), second class (once).